

Training results

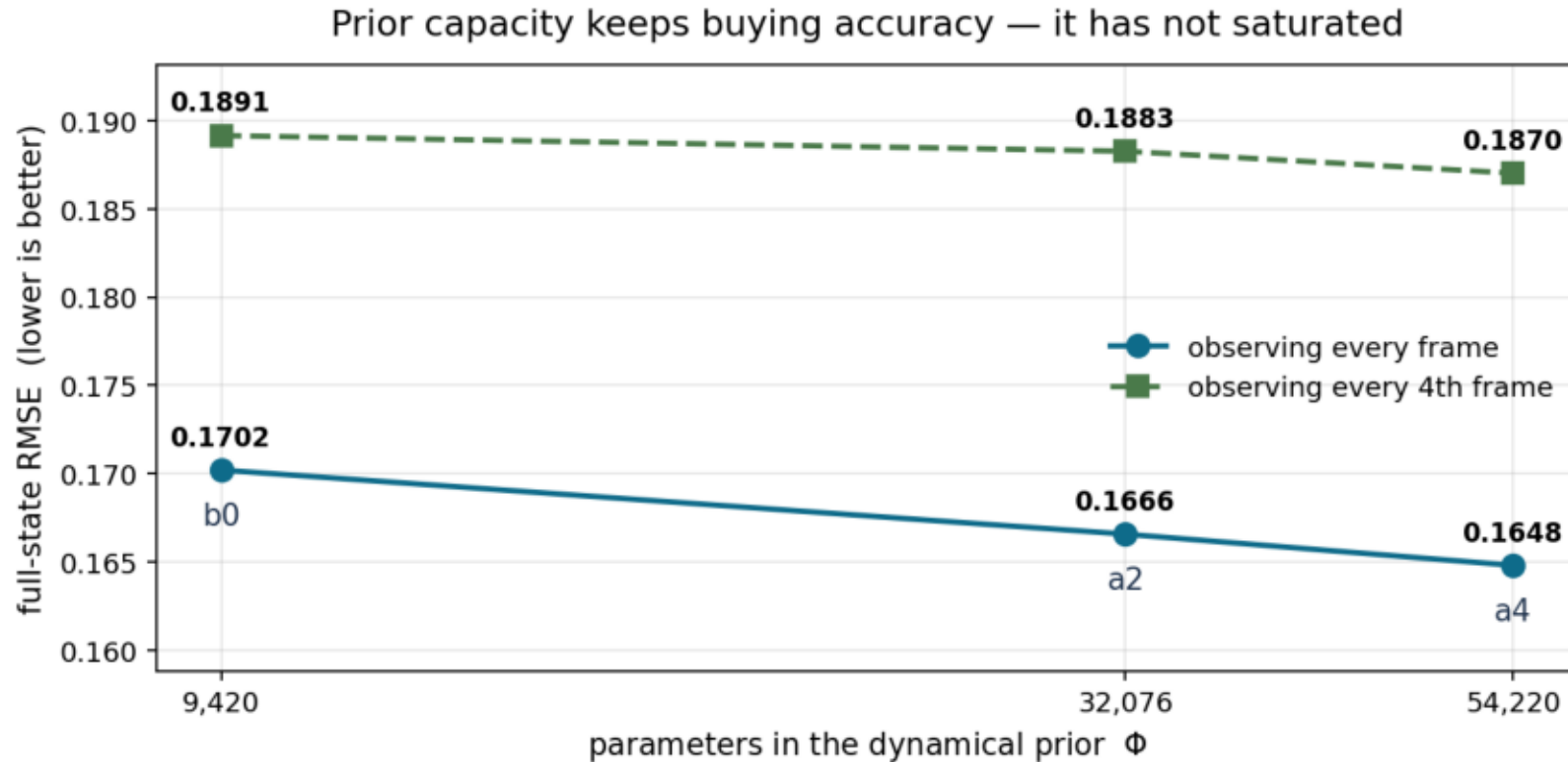
Four configurations of the reconstruction, 5 ensemble members, all on the same 7 held-out test days

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What has been trained

- b0 prior 9,420 params (hidden 32, kt 3) — MODEL OF RECORD
 - full-state RMSE 0.1702 (k=1) / 0.1891 (k=4); best epoch 145 of 150, still improving when it stopped
- a2 prior 32,076 params (hidden 64, kt 5)
 - RMSE 0.1666 / 0.1883 — 2.1% better than b0 at k=1, for 3.4x the prior and 2.15x the inference time (3.66 -> 7.86 ms/frame, same 4-core CPU)
- a4 prior 54,220 params (hidden 96, kt 5)
 - RMSE 0.1648 / 0.1870 — the best accuracy of the three, but its training destabilised at epoch 52 and never recovered
- ml5 b0's architecture, trained on the Gaussian NLL, 5 members
 - RMSE 0.1732 ± 0.0005 — +1.8% against b0, in exchange for a calibrated per-cell uncertainty

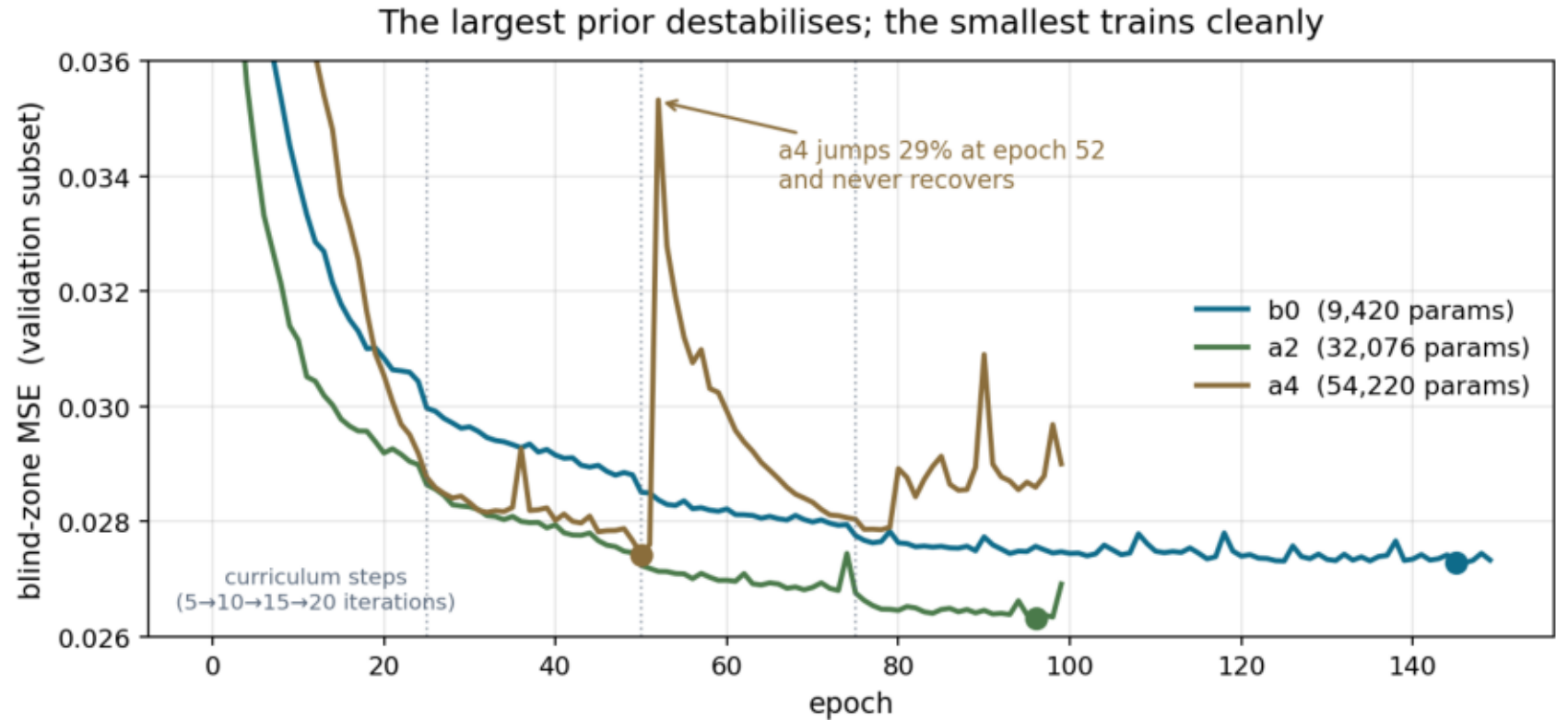
Prior capacity has not saturated



7 held-out test days, one evaluation convention for every point. From 9,420 to 54,220 prior parameters (5.8x) the full-state RMSE falls 3.2% at $k=1$. The curve is still descending, so the capacity question is not settled by accuracy alone — see the training curves for why the largest configuration is nevertheless not the one used.

- From 9,420 to 54,220 prior parameters — 5.8x — the full-state RMSE falls 3.2%, and it is still descending. Accuracy alone would argue for the largest prior.

But the largest prior does not train reliably

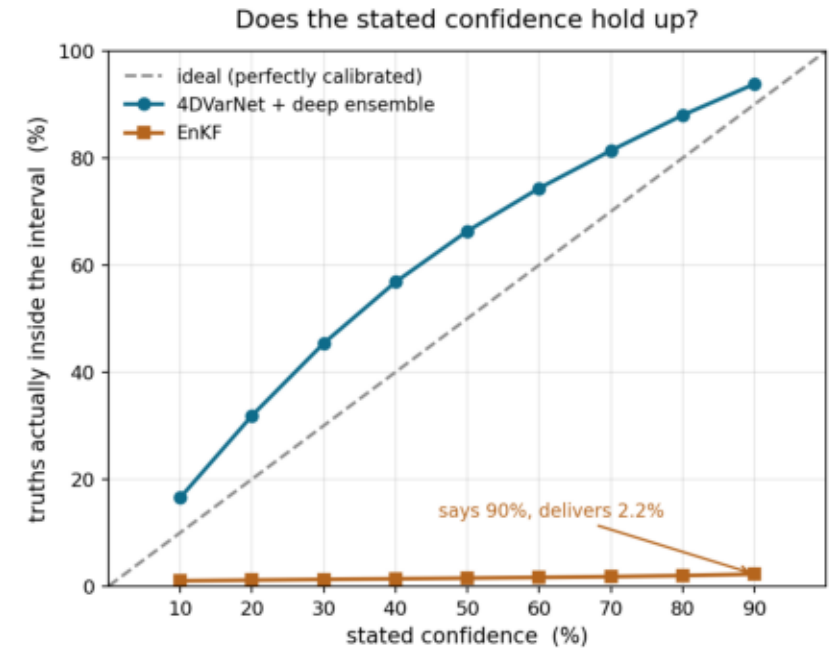


Dots mark each run's best epoch, which is the checkpoint every reported number comes from. b0 improves smoothly to epoch 145. a4's best is epoch 50; at epoch 52 it jumps from 0.0274 to 0.0353 and spends the remaining 47 epochs above where it started, so the 16 epochs added later changed nothing. The jump lands two epochs after the curriculum raises the solver from 10 to 15 iterations — a longer gradient path through a larger prior, which --clip-grad 1.0 did not contain. Validation subset, used for model selection only, never reported as a result.

- a4's best epoch is 50. At epoch 52 its blind-zone error jumps 29% and stays above its own starting point for the remaining 47 epochs — the 16 epochs added to finish the run changed nothing. b0 trains smoothly throughout.

What the NLL objective bought: a usable uncertainty

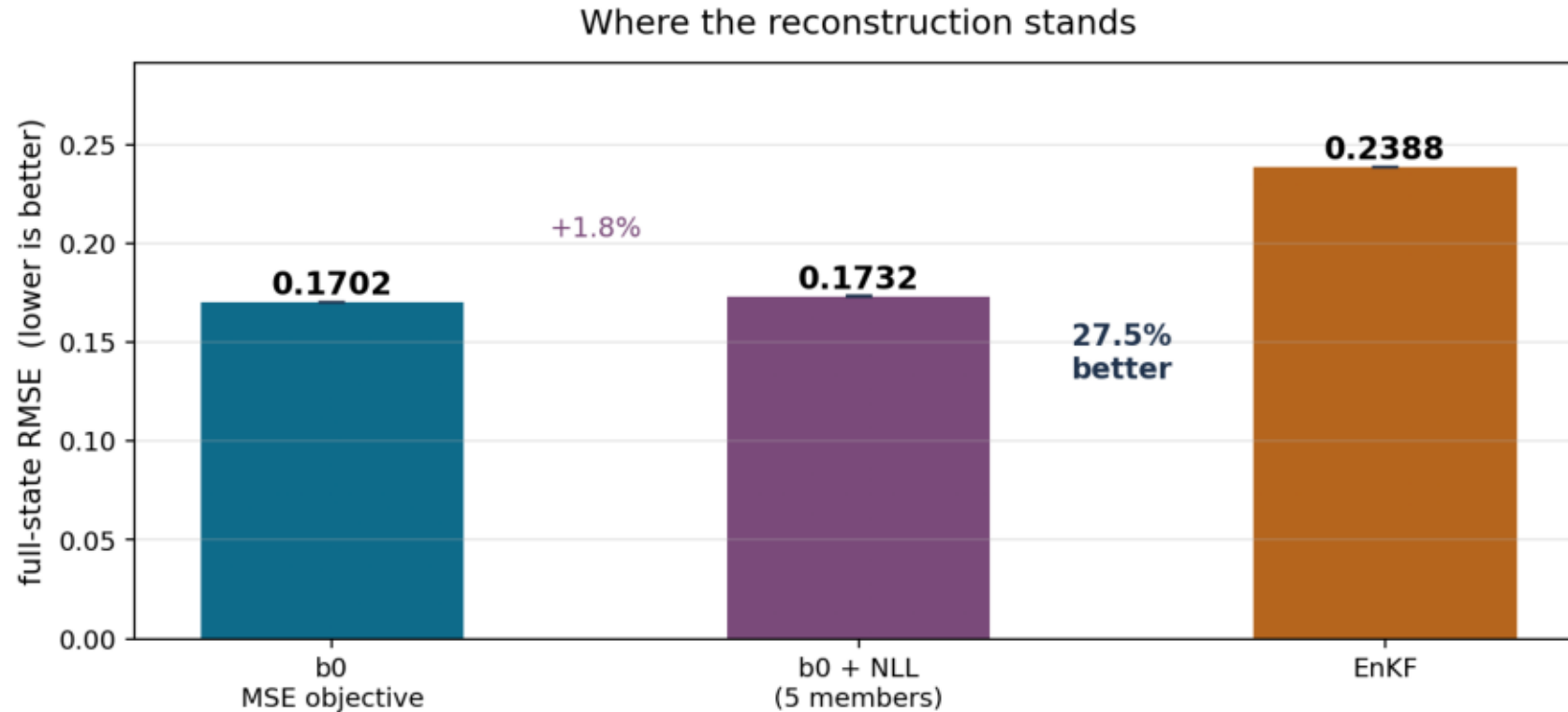
- spread / skill — ideal is 1.0
 - ours 0.87 overall, 0.99 in the blind cells · EnKF 0.011
- 90% interval actually contains — ideal is 90%
 - ours 93.8% · EnKF 2.2%
- does sigma know WHERE it is unsure?
 - ours: 0.1960 in blind cells against 0.1177 in observed ones (1.67x)
 - EnKF: 0.0025 against 0.0025 — identical, it cannot tell them apart



Protocol from Lakshminarayanan et al. App. A.2. A point on the diagonal means the method's stated confidence is exactly right. Above it is conservative (intervals wider than needed); below it is overconfident. The EnKF's curve is nearly flat: widening its interval 9x moves coverage only from 1.00% to 2.24%, so its spread barely tracks its error and no rescaling would fix it.

- The blind cells — the ones this exists for — are close to ideal at 0.99. The observed cells are over-confident at 0.71, so we are not uniformly on the safe side; the EnKF is over-confident everywhere by roughly 90x.
- Structurally the matched comparison is against the members' disagreement alone: 0.0890 against the EnKF's 0.0025.

Where the reconstruction stands



Same 7 test days and the same convention for all three. The error bar on the middle bar is the spread across the 5 independently initialised members (std 0.0005, i.e. 0.3% — they converge to almost the same solution). The EnKF's configuration is the original project's and was not retuned.

- Against the EnKF baseline at $k=1$: b0 is 28.7% better, and the NLL variant 27.5% better while also reporting a per-cell uncertainty. The 5 members agree to within 0.3%.